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A Project Proposal

on

"EmerG-Net"

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Abstract

This project introduces "EmerG-Net," a novel framework designed to identify and quantify the future impact potential of emerging research communities within scholarly literature. Moving beyond retrospective metrics like citation counts, EmerG-Net leverages temporal heterogeneous graph modeling and a synthetic "Boom Score" to detect nascent, high-potential research clusters before they reach scientific saturation. We propose a Deep Residual Graph Neural Network (DrGNN) architecture, integrated with BERT-based (SciBERT) contextual embeddings and graph-based features, to capture both the semantic and structural dynamics of research. The Boom Score will be a multi-dimensional metric, quantifying awakening time, structural growth, and topical shift, and will be optimized using a Learning to Rank strategy. This framework aims to provide a predictive tool for early trend detection, enabling proactive investment in groundbreaking research and democratizing the evaluation of scientific innovation.

Keywords: Emerging Research Communities, Scholarly Knowledge Graphs, Boom Score, Graph Neural Networks, Temporal Modeling, Scientific Impact Prediction

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Acronyms/Abbreviations

SKG: Scholarly Knowledge Graphs

SB: Sleeping Beauty

BERT: Bidirectional Encoder Representations from Transformers

GNN: Graph Neural Network

GCN: Graph Convolutional Network

GResNet: Graph Residual Network

DrGNN: Deep Residual Graph Neural Network

LTR: Learning To Rank

Chapter 1 Introduction

The identification of emerging research communities within the vast, rapidly expanding landscape of scholarly literature is a critical frontier in bibliometrics. As global scientific output continues to accelerate, traditional retrospective indicators such as raw citation counts or **H-indices** have proven inadequate for the early-stage detection of revolutionary shifts. These metrics often lag years behind the actual point of intellectual divergence, failing to capture the nascent momentum of topics that eventually redefine entire disciplines. For example, the structural "boom" in the research graph following the introduction of the Transformer architecture preceded its massive citation explosion. This project proposes a framework for temporal heterogeneous graph modeling centered on a synthetic "**Boom Score**". This score is designed to quantify the future potential of emerging communities groups of papers in early formation before they reach scientific saturation.

1.1 Background

Scientific citation graphs have evolved from static lists into complex, multidimensional Scholarly Knowledge Graphs(SKGs) that map the intricate, time-dependent interactions between authors, institutions, and evolving concepts. Unlike traditional social networks, these are inherently temporal and intransitive; for instance, if Paper A cites Paper B, and Paper B cites Paper C, a connection between A and C only exists if the chronological sequence permits a logical flow of information(Holme and Saramäki 2012). To accurately model these dynamics, researchers must organize data into discrete graph snapshots, $G = \{G_1, G_2, \dots, G_n\}$, which represent the state of the network at specific intervals. This structural framing is essential for capturing the fluidity of scientific discourse and the shifting importance of specific nodes over time.

While most research follows a predictable trajectory peaking shortly after publication before fading, some topics exhibit the "Sleeping Beauty" (SB) phenomenon, characterized by a prolonged hibernation period followed by a sudden, dramatic spike in recognition (Raan 2004; Ke et al. 2015). These delayed breakthroughs often occur when a "Prince" paper from a different discipline "awakens" the dormant idea, bridging a gap that traditional citation models frequently overlook. To effectively predict and identify these non-linear trends, we must move beyond the restrictive "bag-of-words" approach, which ignores semantic con-

text. Instead, by utilizing deep learning architectures like Transformers(BERT), we can create contextual representations of document text that simultaneously preserve positional, structural, and temporal context(Devlin et al. 2019), providing a holistic view of how ideas hibernate and eventually flourish.

1.2 Problem Statement

Existing community detection methods face three critical challenges when applied to emerging research:

1. The Breakdown of Static Modularity: In growing networks, traditional modularity optimization tends to cluster nodes based on their appearance time (age) rather than their true structural affinity, leading to fragmented or time-restricted communities (Medo et al. 2019).
2. Structural and Semantic Disconnect: Models that rely solely on citation structure neglect the rich topical information in text, while topic models often ignore the “credit units” attributed via citations(Huang et al. 2018).
3. The Absence of a Predictive Boom Metric: Existing graph-based frameworks provide no quantitative measure of a research community’s future impact potential. Current metrics such as citation counts, modularity scores, and disruption indices are either retrospective or evaluate individual papers in isolation(Ke et al. 2015; Patel and Javed Ali 2022), offering no mechanism to assess whether an emerging cluster of papers collectively signals an impending paradigm shift before it occurs.

1.3 Objectives

Our project aim to fulfill the following tasks:

1. Using Deep Residual Graph Neural Network (DrGNN) to ensure raw feature information from early layers is preserved, maintaining representation power in deep layers to combat oversmoothing(J. Zhang and Meng 2019).
2. Predictive Modeling of the **”Boom Score”**: Develop a synthetic metric quantifying a community’s future impact through three dimensions: Awakening Time (t_a) (deviation from linear citation growth), Structural Growth (density of time-respecting paths), and Topical Shift (speed of new keyword emergence).
3. Integrating Textual and Structural Data: Combine BERT-based(SciBERT) (Beltagy, Lo, and Cohan 2019) contextual embeddings with graph-based features to create a

unified representation that captures both the semantic and structural dynamics of research communities using node2vec(Grover and Leskovec 2016) and Graph Neural Networks(Kipf and Welling 2016).

4. Early Detection of Emerging Communities: Implement a framework that identifies and ranks emerging research communities based on their predicted Boom Scores, enabling researchers to discover promising areas of study before they reach mainstream recognition.

1.4 Motivation and Significance

The modern research landscape is defined by an exponential surge in scientific output, creating a volume of information that often obscures high-potential "rising stars" and nascent trends. Traditional evaluation metrics, such as citation counts and H-indices, are inherently retrospective and fail to capture the immediate momentum of intellectual shifts. This creates a significant "latency gap," where revolutionary discoveries often referred to as "Sleeping Beauties" may hibernate for years before receiving recognition (Raan 2004; Ke et al. 2015). To address this, there is a critical need for predictive tools that move beyond lagging indicators to identify research topics likely to redefine their disciplines before they reach saturation. Such a framework would allow funding agencies and institutions to move from reactive tracking to proactive investment, maximizing the growth of scientific knowledge by supporting breakthroughs in their infancy.

To bridge this gap, this project aims to introduce a holistic modeling approach that integrates topological momentum with deep semantic analysis. By utilizing Deep Residual Graph Neural Networks (DrGNN)(J. Zhang and Meng 2019), the framework overcomes technical hurdles like oversmoothing to model complex dependencies within vast scholarly knowledge graphs. Unlike conventional methods that rely on prestige or raw popularity, this system employs the Graph structure of citations and SciBERT(Beltagy, Lo, and Co-han 2019) embeddings to differentiate between incremental work and research that truly eclipses its predecessors. By focusing on a paper's objective structural position and its potential to redirect a field, this approach democratizes research evaluation and provides a data-driven lens for detecting the next major explosion in innovation, such as the structural convergence that preceded the Transformer architecture.

Chapter 2 Related Work

2.1 Citation Networks as Scholarly Knowledge Graphs

The formal study of scientific collaboration and citation networks was pioneered by (Newman 2001), who demonstrated that co-authorship graphs exhibit small-world properties, establishing that scientific knowledge diffuses through a surprisingly compact network of intermediary connections. The dynamic nature of these networks was later formalized by (Holme and Saramäki 2012), who proved that static projections of temporal graphs discard critical ordering information, since time-respecting paths are the only valid routes for information propagation and do not necessarily coincide with paths in the static graph. This temporal intransitivity is a foundational concern for this project, motivating the snapshot-based graph formulation used to model the evolving state of a citation network. At the macro level, (Qian, Liu, and Sheng 2020) used hierarchical non-negative matrix factorization to show that scientific revolutions in artificial intelligence were driven by large-scale topic migrations across adjacent subfields converging on a shared vocabulary, a finding that shapes how this project conceptualizes community-level booms as cross-disciplinary convergence events rather than isolated citation spikes.

2.2 Community Detection in Static and Dynamic Networks

Community detection provides the structural backbone for identifying clusters of related papers. (Blondel et al. 2008) introduced the Louvain method, a two-phase heuristic for modularity maximization running in near-linear time, making community detection tractable for large scholarly graphs. However, (Fortunato and Barthélemy 2007) proved that modularity optimization has a fundamental resolution limit, as it cannot detect communities smaller than a scale set by total network size, causing small but distinct emerging clusters to be incorrectly merged, which is a critical deficiency for early-stage discovery. (Traag, Waltman, and Eck 2019) addressed this with the Leiden algorithm, which adds a refinement phase guaranteeing community connectivity after each iteration. (Medo et al. 2019) further demonstrated that modularity breakdown in growing networks is not fundamental but rather a consequence of a mismatch between the observation window and the network’s intrinsic aging timescale, suggesting that community structure in citation networks can be reliably recovered when the analysis window is properly calibrated. The canonical survey by (Rossetti and Cazabet 2018) formalized the full vocabulary of community lifecycle events

including birth, death, merge, split, growth, and contraction, and the tracking approach of (Greene, Doyle, and Cunningham 2010) provides a practical set-overlap mechanism for linking communities across temporal snapshots. More recently, (Lin et al. 2026) and (Sahu 2024) extended Leiden to support efficient incremental updates on continuously evolving graphs. Looking forward, (Jung and Segev 2014) framed community-level forecasting as an actionable research investment signal, demonstrating that node and link prediction methods can identify forming and disbanding communities multiple years in advance.

2.3 Semantic Representation of Scientific Text

Structural graph features alone are insufficient for capturing the intellectual content of a research community. Early approaches relied on topic models such as Latent Dirichlet Allocation (Jelodar et al. 2019), which represent documents as mixtures of latent topics but discard the contextual and positional information crucial for distinguishing semantically related papers. (Devlin et al. 2019) addressed this with BERT, a bidirectional Transformer producing contextual representations that simultaneously encode semantic, positional, and syntactic information. This was extended to the scientific domain by (Beltagy, Lo, and Cohan 2019) with SciBERT, pre-trained on a large multi-domain corpus of scientific publications and achieving state-of-the-art results on scientific NLP tasks; SciBERT serves as the primary embedding backbone in this project. (Grootendorst 2022) pushed this further by replacing bag-of-words topic modeling entirely with contextual Transformer embeddings clustered and labeled through a class-based TF-IDF procedure, producing topics that better track the vocabulary of emerging research clusters. For detecting the emergence of new topics, (Kleinberg n.d.) established that nascent discourse is best identified by the rate of change in keyword frequency rather than raw occurrence counts, directly motivating the topical shift component of the proposed Boom Score. (Huang et al. 2018) further demonstrated that structural position and topical relevance are complementary signals, showing that jointly modeling both yields superior identification of influential and rising papers over citation structure alone.

2.4 Deep Graph Neural Networks and Oversmoothing

Graph Neural Networks have become the dominant paradigm for learning over relational data. (Kipf and Welling 2016) introduced the Graph Convolutional Network, which aggregates features from local neighborhoods through a spectral approximation, while (Hamil-

ton, Ying, and Leskovec 2017) extended this with GraphSAGE, an inductive framework that generates embeddings for unseen nodes as the graph grows, a critical property for a continuously expanding citation network. (Grover and Leskovec 2016) proposed node2vec, learning structural embeddings via biased random walks that interpolate between local and global graph context. A well-known pathology of deep GNNs is oversmoothing, whereby repeated neighborhood aggregation causes node representations to converge to indistinguishable vectors. (Li, Müller, et al. 2019) introduced residual and dense connections into GCN architectures to mitigate this, enabling significantly deeper networks, while (Li, Xiong, et al. 2020) extended this with differentiable generalized aggregation and pre-activation residual design. (J. Zhang and Meng 2019) proposed GResNet, which constructs highways carrying raw input features to all layers simultaneously, preventing the suspended animation failure mode where deep models become unresponsive to training data. The DrGNN architecture of (Zheng et al. 2024), which this project aims to directly adopt, reframes oversmoothing as a dimensional collapse problem and resolves it through topology-preserving residual connections, achieving state-of-the-art results on deep graph representation benchmarks.

2.5 Bibliometric Indicators of Impact and Delayed Recognition

Retrospective citation metrics systematically fail early-stage discovery. (Raan 2004) identified "Sleeping Beauties" papers dormant for years before sudden recognition exposing the inability of short-window metrics to capture non-linear impact dynamics. (Ke et al. 2015) formalized this across 22 million papers with the Beauty Coefficient, showing delayed recognition is a continuous spectrum rather than a rare exception. Beyond timing, (Patel and Javed Ali 2022)'s Disruption Index distinguishes knowledge-consolidating papers from field-redirecting ones, directly informing the Boom Score's structural growth dimension. (F. Zhang and Wu 2024) further demonstrates that multi-domain datasets require domain-aware modeling, a principle this project adopts through community-level feature aggregation.

Chapter 3 Design and Implementation

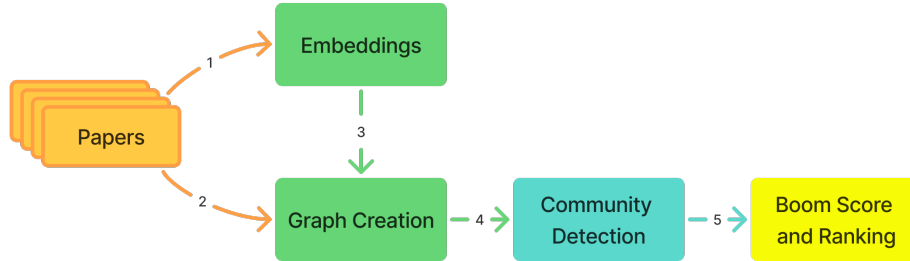


Figure 1: High-level System Architecture Overview

3.1 Data Collection

For the data collection phase, we will utilize the **OpenAlex**(OpenAlex Team 2022) dataset, a comprehensive and fully open bibliographic catalog of the global research ecosystem. OpenAlex is uniquely suited for our objective of discovering emerging communities in a graph because it provides a massive, interconnected web of scholarly entities and their relationships. By treating this data as a heterogeneous network, we can identify clusters of research activity as they form.

The following key attributes from OpenAlex will be extracted to construct the nodes and edges of our graph:

1. **Openalex ID**
2. **DOI**
3. **title**
4. **publication date**
5. **authorships**
6. **fwci**
7. **cited by count**
8. **topics**
9. **referenced works**
10. **inverted index of Abstract**

3.2 Graph Construction

This graph will be implemented using NetworkX(NetworkX Developers 2008) for rapid prototyping and small-scale analysis or igraph(igraph Developers 2006) for handling larger datasets with efficient querying capabilities and effective implementation of foundational traditional community detection algorithms. Gephi will be used for visualizing the graph structure and community clusters, providing insights into the overall topology and emerging patterns.

3.2.1 Citation Graph

The citation graph represents papers as nodes and citation links as directed edges, where an edge from Paper A to Paper B indicates that A cites B.

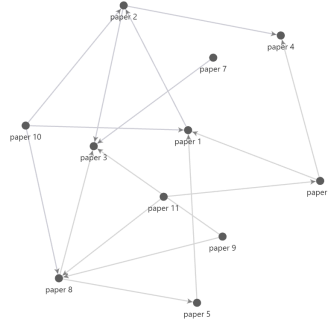


Figure 2: Example citation graph with dummy nodes

3.2.2 Paper-Author Graphs

Paper-Author Graphs will extend the citation graph by incorporating additional author information: author networks (edges between co-authors and edges from papers to their authors). This multi-layered graph provides a holistic view of academic connections, enabling more nuanced retrieval by linking papers through authors and beyond direct citations.

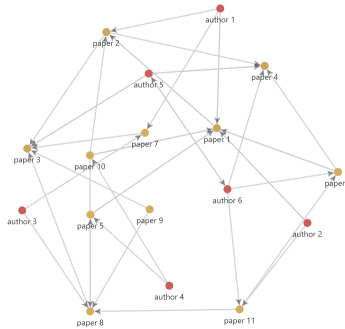


Figure 3: Example paper author graph with dummy nodes

3.3 Community Detection

To identify emerging research communities, we will apply a combination of traditional and advanced community detection algorithms. Traditional methods such as Leiden(Traag, Waltman, and Eck 2019) and Infomap(Smiljanić et al. 2023) will be used to establish baseline community structures based on modularity optimization. These algorithms will help us understand the initial clustering of papers based on citation patterns.

3.4 Topic Extraction

After communities have been identified every community will be assigned a topic as a aggregated vote from all nodes in the community. The topics from each paper will be extracted using BertTopic(Grootendorst 2022) or KeyBert(Issa et al. 2023).

3.5 Embeddings

3.5.1 Semantic Embeddings of Papers

Semantic representations will be generated for the text of each paper (title and abstract) to fully capture its content and context. SciBERT(Beltagy, Lo, and Cohan 2019), a transformer model fine-tuned on scientific texts, will process the full text in segments (e.g., 512-token chunks) due to its input constraints, producing contextual embeddings (768-dimensional vectors) for each segment. These will be averaged across segments to create a single content-based embedding per paper, reflecting its scientific meaning.

3.5.2 Embeddings of Authors

Author embeddings will be generated by aggregating the embeddings of their published papers, their researcher statistics like H-index, and their co-authorship network features. This will create a comprehensive representation of each author’s influence and research focus.

We may also use a densely connected feedforward neural network to further refine the author embeddings.

3.5.3 Graph-Based Embeddings

After obtaining the SciBERT embeddings, we will use node2vec(Grover and Leskovec 2016) to further transform the embeddings to include the structural nuances. This will serve as our baseline for the graph-based models, capturing both the semantic content and the structural position of each paper within the citation network.

For GNNs, we will utilize the foundational GCN(Kipf and Welling 2016) and DrGNN architecture(J. Zhang and Meng 2019) to create a unified representation that integrates both the semantic and structural dynamics of research communities.

3.6 The Ranking of Communities

For the ranking of communities, we will compute a **”Boom Score”** for each community using dense neural layers. Rather than predicting a fixed growth value, the model will be trained using a **Learning to Rank (LTR)** strategy. This approach focuses on the relative potential of communities, identifying which topics are likely to outperform others within the same academic domain.

3.6.1 Training Loop

The model will be trained using a **Pairwise Ranking** objective, specifically a RankNet-inspired framework, to optimize the structural and semantic ordering of communities:

- **Input Representation:** Each community C_i is represented by a feature vector x_i that concatenates structural graph metrics (from Leiden/Infomap) with semantic embeddings (from BERTopic).
- **Pairwise Objective:** During training, the model f_θ processes pairs of communities (C_i, C_j) sampled from the same time snapshot. If Community C_i historically showed higher citation growth in subsequent OpenAlex snapshots than C_j , the model is penalized if $f_\theta(x_i) < f_\theta(x_j)$.
- **Optimization:** We utilize a **Binary Cross-Entropy loss** over the score differences, forcing the model to maximize the margin between ”booming” and ”stable” communities.

By adopting a ranking-based approach, the framework remains robust to the high variance in citation scales across different scientific fields, ensuring that ”breakout” topics are prioritized regardless of their niche.

Chapter 4 System Requirement Specification

4.1 Software Specifications

4.1.1 Front-End Technologies

- Core UI Library: React.js
- Next.js (React Framework)
- Plotly (for interactive visualizations)

4.1.2 Back-End Technologies

- Python (primary programming language)
- FastAPI (for building scalable and high-performance APIs)

4.1.3 Database Systems

- Faiss Vector Database
- Neo4j Graph Database

4.1.4 Machine Learning and AI Frameworks

- PyTorch (for training and fine-tuning deep learning models)
- Pytorch Geometric (for GNNs)
- igraph (for graph analysis)

4.2 Hardware Specifications

4.2.1 Minimum Hardware Requirements

- Processor: Intel Core i7 (or equivalent) with at least 8 cores
- RAM: 16GB (recommended: 32GB for handling large embeddings)
- Storage: SSD with at least 1TB (recommended: NVMe SSD for fast retrieval)
- GPU: NVIDIA RTX 3060 (or higher) for model inference and training

Chapter 5 Project Planning and Scheduling

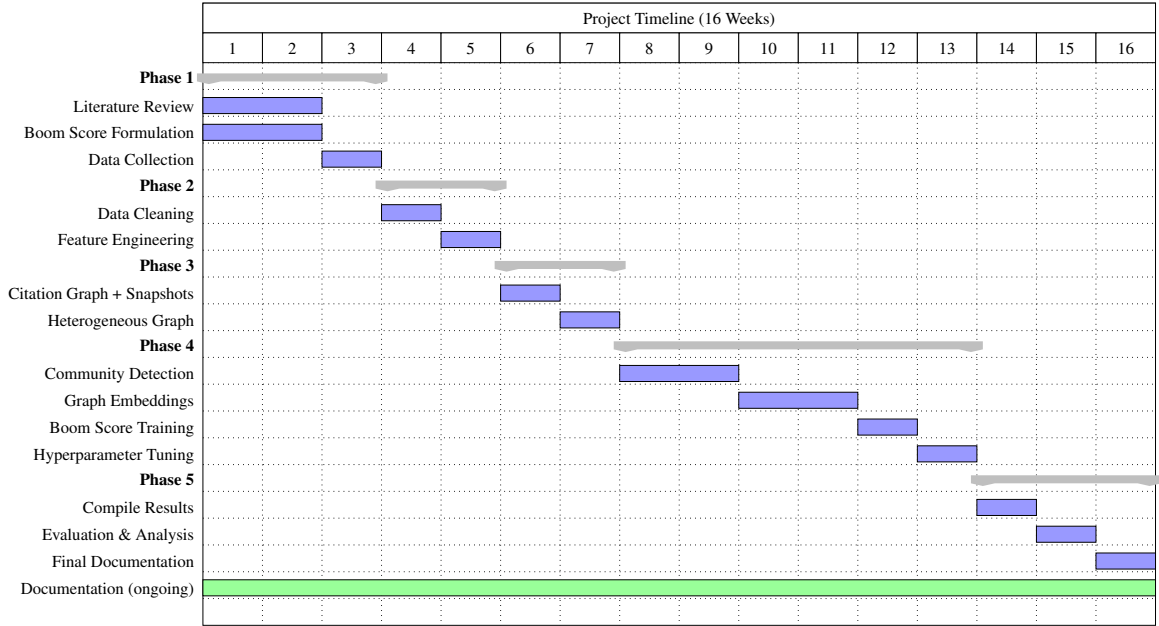


Figure 4: Project Gantt Chart — EmerG-Net development timeline across 16 weeks. Color coding: ■ Phase groups, ■ Task bars, ■ Ongoing documentation.

Phase Descriptions:

- **Phase 1 Foundation (Weeks 1–3):** Literature review and finalization of the Boom Score formulation, followed by data collection via scholarly APIs (OpenAlex).
- **Phase 2 Data Pipeline (Weeks 4–5):** Deep cleaning, deduplication, and schema normalization of raw data, followed by feature engineering including SciBERT embeddings, author statistics, and keyword extraction.
- **Phase 3 Graph Construction (Weeks 6–7):** Construction of temporal citation graph snapshots and extension to a heterogeneous paper-author graph, with structural validation.
- **Phase 4 Modeling and Experimentation (Weeks 8–13):** The core experimental phase, covering community detection baselines (Leiden, Infomap), graph embedding experiments (node2vec, GCN, DrGNN), Boom Score supervised training, and hyperparameter tuning with ablation studies.
- **Phase 5 Results and Wrap-up (Weeks 14–16):** Compilation of results and visualizations, evaluation and failure analysis, and final project documentation and presentation preparation.
- **Documentation (Ongoing, Weeks 1–16):** Incremental documentation maintained throughout all phases to avoid end-stage bottlenecks.

Chapter 6 Expected Outcome

The expected outcome of this project is the development of a novel predictive metric, termed the “Boom Score,” designed to quantify the future impact potential of emerging research communities before they achieve widespread recognition. Unlike traditional retrospective indicators such as citation counts or H-index, the Boom Score will integrate multiple forward-looking signals, including temporal citation dynamics (awakening time), structural growth patterns within the graph, and semantic evolution of research topics. This metric aims to capture early-stage momentum and identify non-linear trends such as “Sleeping Beauty” phenomena, enabling the detection of high-potential research clusters at their formative stages.

In addition to the metric itself, the project will deliver a reusable and extensible framework for modeling and analyzing temporal heterogeneous graphs. This framework will unify semantic embeddings derived from transformer-based models with structural representations learned through graph-based techniques such as node2vec and Graph Neural Networks. Designed with modularity and scalability in mind, the system can be adapted beyond scholarly citation networks to other domains such as social media, innovation networks, and knowledge graphs. Together, the Boom Score and the supporting framework will provide a general-purpose tool for early trend detection, offering both practical utility and a foundation for future research in predictive graph analytics.

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